University Final Project

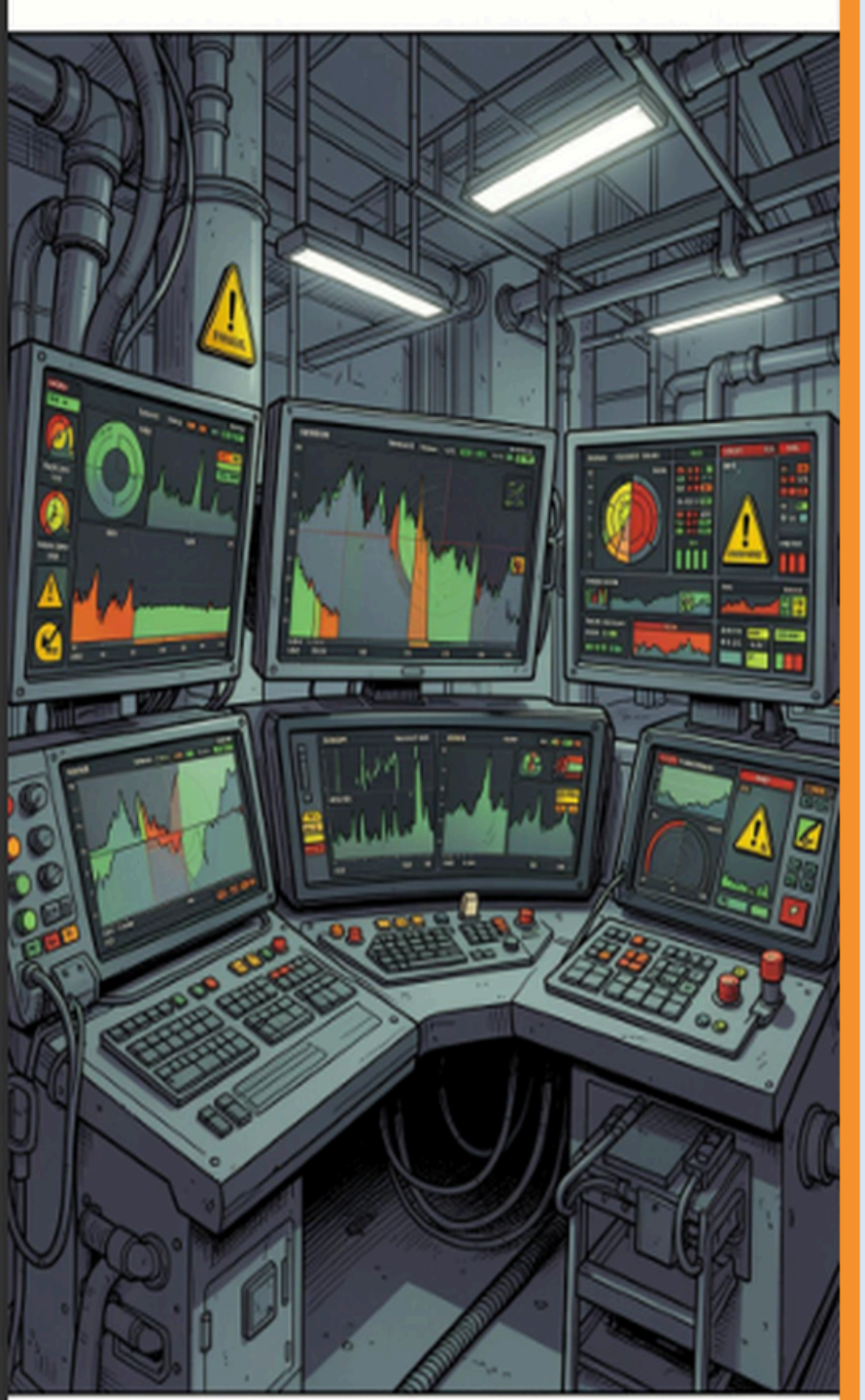
SmartFactory – Industrial Production Analytics Platform

Applying advanced data structures and algorithms to smart manufacturing.

Machines · Production Lines · Raw Materials · Quality Control · Maintenance · Analytics



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The Industrial Data Challenge

Modern factories generate massive volumes of sensor telemetry and operational reports. Without efficient indexing and search, critical fault data stays buried.

Noisy, High-Volume Telemetry

Sensor streams create data too large for linear scanning during fault investigation.

Unplanned Downtime

Machine failures demand fast retrieval of fault records to minimize production loss.

Unindexed Historical Reports

Maintenance logs grow continuously, making repeated queries increasingly inefficient.

Data Entry Errors

Manual machine ID and telemetry entry introduces typos that corrupt records.

M1 / CO1 — Find the right machine signal fast

KMP: Search machine IDs, production codes, and fault descriptions.

Z-Function: Identify repeated maintenance-report patterns.

Rabin–Karp: Search production and inventory codes.

Aho–Corasick: Detect multiple machine-fault keywords.

Example: “Machine M102 has overheating fault” → detect Machine ID + Fault Keyword.



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M2 / CO2 — Index years of production knowledge

Document Similarity

Suffix Arrays and SA-IS index large production and maintenance reports.
LCP/Kasai surfaces repeated fault descriptions.
Suffix Trees and Suffix Automata support substring-based report queries.

Report	Repeated phrase (LCP)
Log 2024-04	**vibration spike detected at spindle motor**
Log 2025-11	**vibration spike detected at spindle motor**

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M3 / CO3 — Correct messy data and optimize choices

Dynamic Programming

Levenshtein and Damerau–Levenshtein correct machine and production entries.

Weighted edit distance prioritizes likely reporting errors.

Matrix-chain multiplication, bitmask DP, and subset DP optimize data operations and machine/activity combinations.

Raw input		Corrected entity
“Machne M102”	→	**“Machine M102”**



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M4 / CO4 — Move materials where capacity allows

Ford–Fulkerson and Edmonds–Karp
Model and maximize raw-material flow.

Dinic's Algorithm
Scales to larger factory-flow networks.

Bipartite matching
Assigns workers to machines or activities.

Max-flow min-cut
Reveals production bottlenecks.

Capacity at each stage · bottleneck highlighted

1 Raw Materials

2 Production Stages

3 Machines

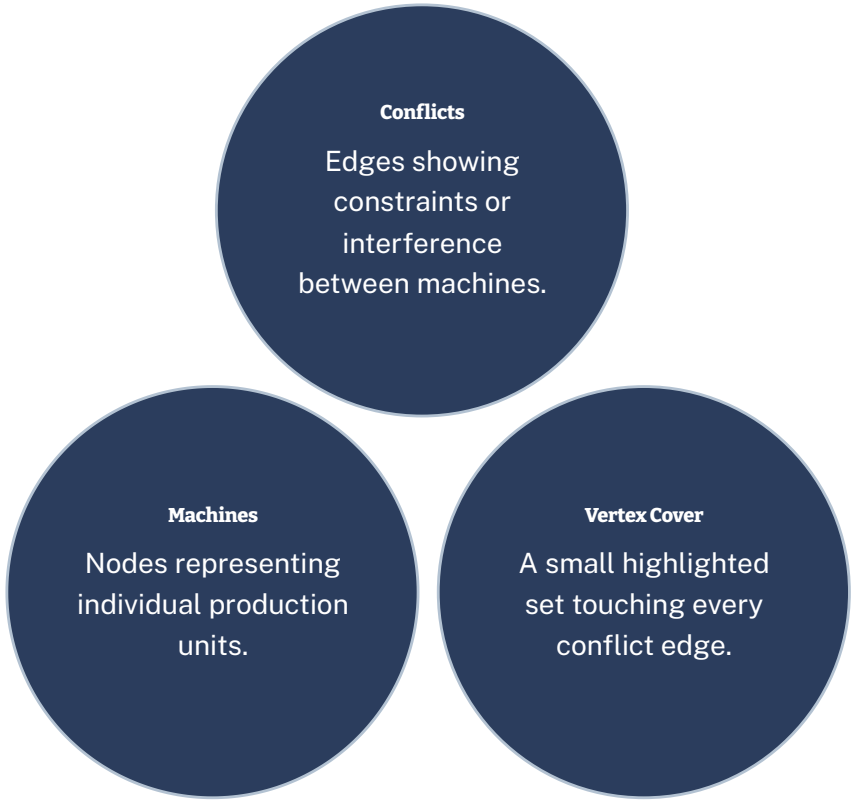
4 Finished Products

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M5 / CO5 — Handle constraints and production conflicts

- SAT/3-SAT represents production constraints.
- 3-SAT → CLIQUE → INDEPENDENT-SET → VERTEX-COVER maps conflict relationships.
- Vertex-Cover 2-Approximation identifies machines or connections that cover conflicts.

Production-conflict graph



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M6 / CO6 — Turn sensor streams into real-time analytics

- **Randomized QuickSort** ranks production orders by priority.
- **Miller–Rabin** supports large numerical computations.
- **Reservoir Sampling** samples continuous machine-data streams.
- **Blelloch Scan, Parallel Reduce, and Brent's Theorem** support scalable aggregation and parallel processing.

Machine Sensors → **Data Stream →** **Parallel Processing →** **Real-Time Analytics**



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Algorithm Complexity & Best Choices

Compare algorithms that solve similar tasks; choose based on input size and workload.

1. STRING SEARCH

KMP: O(n + m)

Z-Function: O(n + m)

Rabin·Karp: expected O(n + m); worst O(nm)

Aho·Corasick: O(P + n + z) after build

BEST: KMP/Z for one pattern; Aho·Corasick for many

2. TEXT INDEXING

Suffix Array (typical): O(n log n)

SA-IS: O(n) construction

Kasai LCP: O(n)

Suffix Tree / Automaton: O(n) build*

BEST: SA-IS for linear-time suffix-array build

3. EDIT DISTANCE & DP

Levenshtein / Damerau / Weighted Edit: O(nm)

Matrix-Chain DP: $O(n^3)$ time, $O(n^2)$ space

Bitmask/Subsets DP: typically O(n · 2·)

BEST: Edit-distance variants tie asymptotically;

choose based on the error model and cost weights

4. NETWORK FLOW

Ford·Fulkerson: O(E·F) for integer capacities

Edmonds·Karp: $O(VE^2)$

Dinic: $O(V^2E)$ general-graph bound

Bipartite matching: O(E·V) with Hopcroft·Karp

BEST: Dinic for general flow; Hopcroft·Karp for large bipartite matching

5. HARD CONSTRAINTS

DPLL / 3-SAT: worst-case exponential, O(2·) search

CLIQUE / Independent Set / Vertex Cover: NP-hard

Reductions: polynomial-time for standard reductions

Vertex Cover 2-approximation: O(V + E)

BEST: 2-approximation when a fast practical near-optimal cover is acceptable

6. RANDOMIZED & PARALLEL

Randomized QuickSort: expected O(n log n); worst $O(n^2)$

Miller·Rabin: $O(k \log^3 N)$ typical arithmetic bound

Reservoir Sampling: O(N) time, O(k) space

Blelloch Scan / Tree Reduction: O(N) work, O(log N) depth

BEST: Reservoir Sampling for streams; parallel scan/reduction for aggregates

Evaluation & Impact

One pipeline, nine measurable benefits

SmartFactory demonstrates how advanced DSA algorithms can improve efficiency, scalability, optimization, and decision-making in modern manufacturing.

Fast search, efficient indexing, data-error correction, optimized processes, resource allocation, bottleneck detection, constraint handling, real-time analytics, and scalable smart manufacturing.

The Complete Pipeline

1. Factory Data
2. String Search
3. Document Indexing
4. Dynamic Programming
5. Network Flow
6. Constraint Optimization
7. Randomized & Parallel Processing
8. Real-Time Factory Analytics

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